

Research Interests	Computer Systems, Data Systems, Distributed Systems, Systems for Machine Learning	
Education	Massachusetts Institute of Technology	2020 – Present
	Doctor of Philosophy Student in Computer Science (PhD)	Cambridge, MA
	• <i>Advisor:</i> Tim Kraska, Data Systems Group	
	University of Toronto	2018 – 2020
	Master of Science in Computer Science (MSc)	Toronto, ON
	• <i>Advisor:</i> Gennady Pekhimenko, EcoSystem Group (Computer Systems and Networks) • <i>Thesis:</i> Habitat: Prediction-guided Hardware Selection for Deep Neural Network Training	
	University of Waterloo	2013 – 2018
	Bachelor of Software Engineering (BSE)	Waterloo, ON
	• 94.65% Cumulative Average, Graduated with Distinction and on the Dean's Honors List	
Publications	Habitat: A Runtime-Based Computational Performance Predictor for Deep Neural Network Training Geoffrey X. Yu , Yubo Gao, Pavel Golikov, and Gennady Pekhimenko USENIX Annual Technical Conference (USENIX ATC), 2021	
	Skyline: Interactive In-Editor Computational Performance Profiling for Deep Neural Network Training Geoffrey X. Yu , Tovi Grossman, and Gennady Pekhimenko ACM Symposium on User Interface Software and Technology (UIST), 2020	
Demonstrations	Skyline: Interactive In-Editor Performance Visualizations and Debugging for DNN Training Geoffrey X. Yu , Tovi Grossman, and Gennady Pekhimenko Conference on Machine Learning and Systems (MLSys), 2020	
	TBD Suite: Benchmarking and Profiling Tools for DNNs Geoffrey X. Yu , Hongyu Zhu, Anand Jayarajan, Bojian Zheng, Abhishek Tiwari, and Gennady Pekhimenko Conference on Machine Learning and Systems (MLSys), 2019	
Teaching Experience	Teaching Assistant	University of Toronto
	CSC 209 – Software Tools and Systems Programming	Fall 2018, Winter 2019
Professional Experience	Facebook	Software Engineering Intern
	Menlo Park, CA	September – December 2017
	• Utilized C++ to develop a new queuing service for the Async Tier—Facebook's asynchronous job execution platform, responsible for handling tens of millions of jobs per minute. • Designed and implemented a mechanism to reschedule failed jobs, requiring development across new and legacy infrastructure services on the Async Tier. • Improved operational efficiency by implementing scripts to facilitate database shard management.	
	Facebook	Software Engineering Intern
	Menlo Park, CA	January – April 2017
	• Implemented robust Java code diagnostics for Nuclide, enabling Java editing support for massive projects with over 60,000 source files. • Engineered and developed Nuclide's Java type resolution mechanism, which was capable of handling source code files with more than 10,000 transitive dependencies. • Added partial rendering to the Nuclide console, increasing the number of entries that could be retained in the user interface by 20 times.	

Apple
Cupertino, CA

Software Engineering Intern
May – August 2016

- [Contributed to Apache Cassandra](#) to improve disaster recovery across data centers.
- Prototyped a Cassandra partition blacklisting mechanism in Java to support nodes in production.
- Improved Cassandra timeout enforcement and introduced custom per-query timeouts
- Enhanced Cassandra error handling to enable error propagation from replica nodes to clients.

Coursera
Mountain View, CA

Software Engineering Intern
September – December 2015

- Developed a near real-time course tracking tool by leveraging Scala, Kafka, and JavaScript with React—providing an aggregate view of over 1700 courses on Coursera.
- Wrote a scheduled job in Scala to validate user role invariants across multiple data stores.
- Utilized Scala and JavaScript with React to build a suite of administrative course editing tools end-to-end, replacing legacy tools that lacked comprehensive validations.

Coursera
Mountain View, CA

Software Engineering Intern
January – April 2015

- Engineered robust course authoring tools using React and Flux that were made available to over 100 Coursera partner institutions.
- Built the front end for an instructor to learner mass email communication tool. This tool has been used to send out over 800,000 emails.

Milq (now Layer 6, acquired by TD Bank Group)
Toronto, ON

Full Stack Software Engineering Intern
May – August 2014

- Developed the tag page—an aggregation of content about a specific hashtag.
- Engineered scalable data API endpoints in Node.js using Redis and MongoDB to deliver matched content to the tag page.
- Developed functionality that fetched, resized, and stored video thumbnails into an Amazon S3 bucket for new and existing content added to Milq.

Awards and Honors

- **NSERC Postgraduate Scholarship – Doctoral (PGS D)** 2020 – 2023
Merit-based doctoral research scholarship awarded by the Government of Canada (63,000 CAD).
- **NSERC Alexander Graham Bell Canada Graduate Scholarship (CGS D)** 2020 – 2023 (declined)
Merit-based doctoral research scholarship awarded by the Government of Canada (105,000 CAD).
- **NSERC Canada Graduate Scholarship – Master’s (CGS M)** 2019 – 2020
Merit-based master’s research scholarship awarded by the Government of Canada (17,500 CAD).
- **Snap Research Scholarship** January 2019
Awarded by Snap Inc. for research quality, technical skills, and communication skills (10,000 USD).
- **Vector Institute Scholarship in Artificial Intelligence** 2018 – 2019
Awarded to top students pursuing AI-related master’s studies in Ontario (17,500 CAD).
- **Queen Elizabeth II Graduate Scholarship in Science and Technology** 2018 – 2019
Merit-based scholarship awarded to students pursuing graduate studies in Ontario (15,000 CAD).
- **Wolfond Fellowship, University of Toronto** 2018 – 2019
Awarded at the University of Toronto for academic achievements (10,000 CAD).
- **Sandford Fleming Foundation Award for Academic Excellence** June 2018
Best graduating academic record in the Software Engineering program at the University of Waterloo.
- **University of Waterloo Faculty of Engineering First in Class Award** May 2017
Highest term average in the Software Engineering program at the University of Waterloo (Fall 2016).
- **Wish Scholarship, University of Waterloo** January 2016
Awarded by the University of Waterloo based on academic merit.

- **University of Waterloo Faculty of Engineering Upper Year Scholarship** May 2015
Awarded by the Faculty of Engineering at the University of Waterloo based on academic merit.
- **Software Engineering Entrance Scholarship, University of Waterloo** September 2013
Awarded to first year Software Engineering students based on academic and extracurricular achievements.
- **University of Waterloo President's Scholarship of Distinction** September 2013
Awarded to undergraduate students with an admissions average of 95% or greater.
- **Toronto District School Board Top Scholar** June 2013
Highest high school graduating average in the Toronto District School Board (Toronto, ON).